



M002 · Final Cube Maze

Topic: Final Cube Maze · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

Final week! Build a 3D cube maze with a door and a lever, then write code that solves it.

1 · Predict 🧠

```
keep going until at goal:
  if wall ahead OR closed door ahead:
    turn right
  else if lever ahead:
    interact
  else:
    move forward
```

Name one maze where this gets stuck.

2 · Trace

```
keep going until at goal:
  if lever ahead:
    interact
  else if wall ahead:
    turn right
  else:
    move forward
```

Agent sees a lever, then open path, then a wall. What does it do each step?

3 · Fill the Blanks

This loop should solve the maze.

```
keep going until at goal:
    if wall ahead:
        agent.turn(_____)
    else:
        agent.move(FORWARD, ____)
```

Word bank: LEFT_TURN · RIGHT_TURN · 1

4 · One Bug 🐛

This must use **interact** **only** when a lever is ahead. It interacts on every step.

```
keep going until at goal:
  interact
  if wall ahead:
    turn right
  else:
    move forward
```

Write the fixed code.

Why was it wrong?

5 · Show Your Work 📷 🎥

Build your final maze:

- **20 × 20 × 20** cube
- more than one floor
- 1 redstone door
- 1 lever the agent flips
- start and finish
- code with `while` + `if/else` + `interact` (use `agent.move(UP, 1)` for floors)

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.